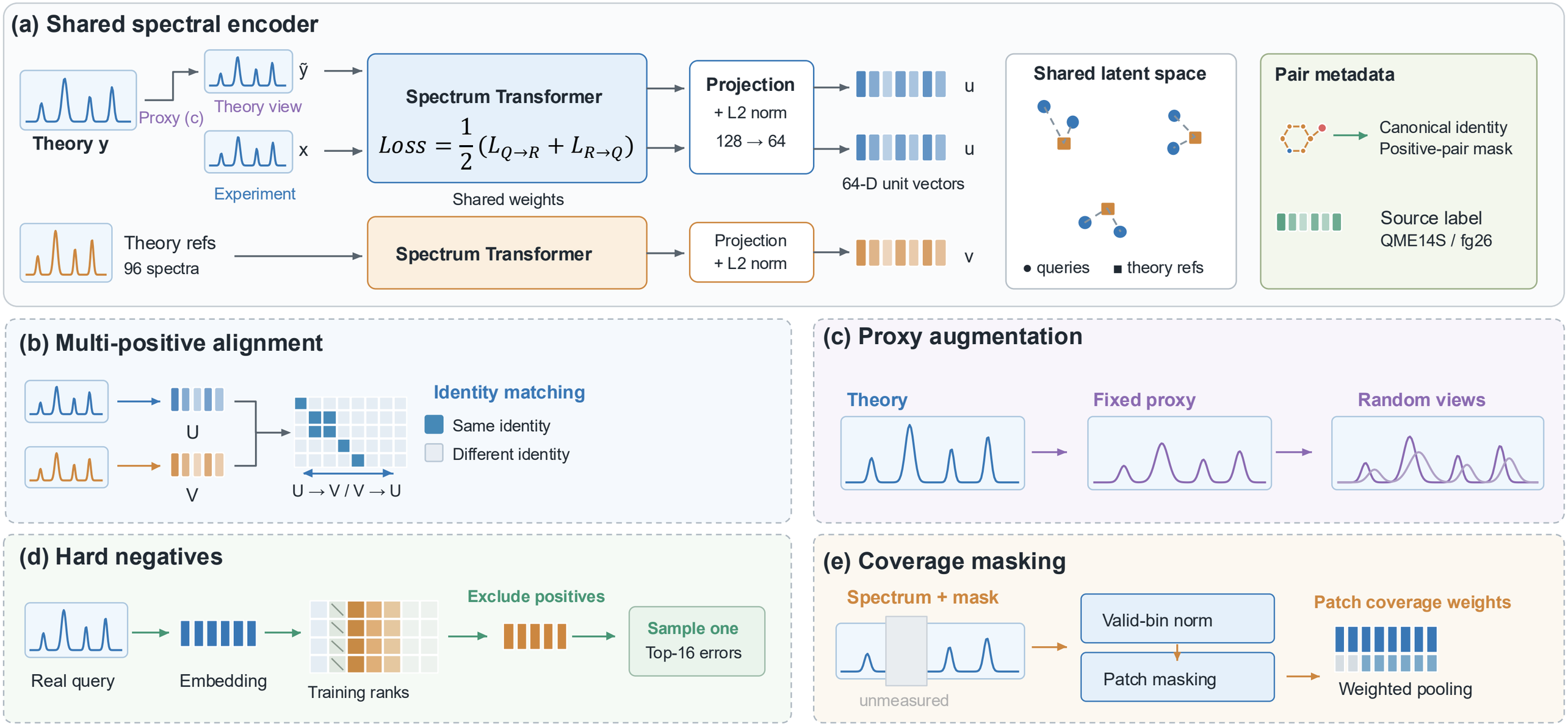

(a) Shared spectral encoder
Theory y
Proxy (c)
Theory view
ỹ
x
Experiment
Spectrum Transformer
$Loss = \frac{1}{2}(L_{Q\to R} + L_{R\to Q})$
Shared weights
Projection
+ L2 norm
128 → 64
u
u
64-D unit vectors
Theory refs
96 spectra
Spectrum Transformer
Projection
+ L2 norm
v
Shared latent space
queries
theory refs
Pair metadata
Canonical identity
Positive-pair mask
Source label
QME14S / fg26
(b) Multi-positive alignment
U
V
U → V / V → U
Identity matching
Same identity
Different identity
(c) Proxy augmentation
Theory
Fixed proxy
Random views
(d) Hard negatives
Real query
Embedding
Training ranks
Exclude positives
Sample one
Top-16 errors
(e) Coverage masking
Spectrum + mask
unmeasured
Valid-bin norm
Patch masking
Patch coverage weights
Weighted pooling